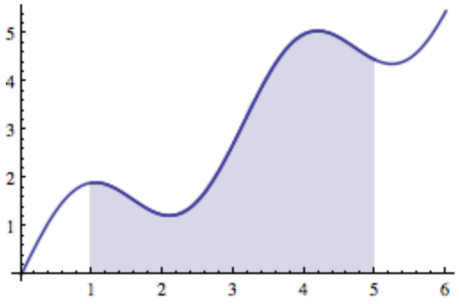
Consider the function $f(x) = x + \sin(2x)$ on the interval $[1,5]$. The integral $\int_1^5 f(x)dx$ is the area under the curve, shown in the figure below:



The exact area is given by

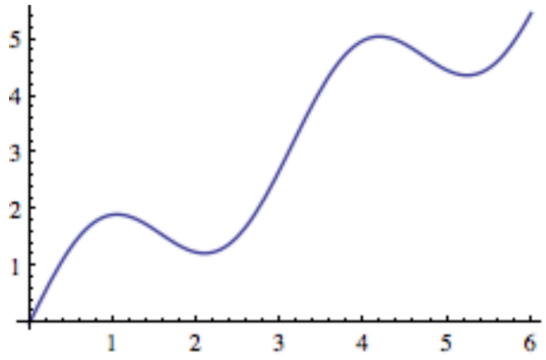
$$\int_1^5 (x+\sin(2x))dx = \left[\frac{x^2}{2} - \frac{1}{2}\cos(2x)\right]_1^5 = \frac{25}{2} - \frac{1}{2}\cos(10) - \frac{1}{2} + \frac{1}{2}\cos(2) \approx 12.211.$$

Calculate each of the following Riemann sums

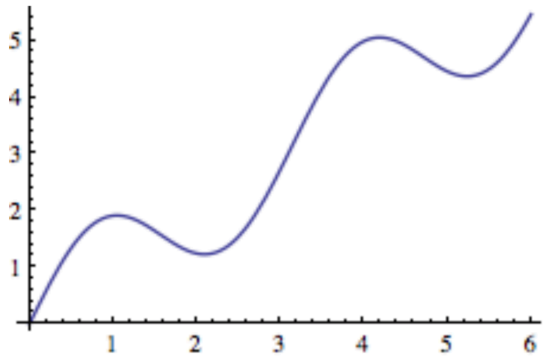
$$\sum_{i=1}^{n} f(x_i^*)(x_i - x_{i-1})$$

to three decimal places and compare your answer to the answer above. Determine if the Riemann sum is an overestimate or an underestimate. Draw the rectangles of your Riemann sum on the provided graph.

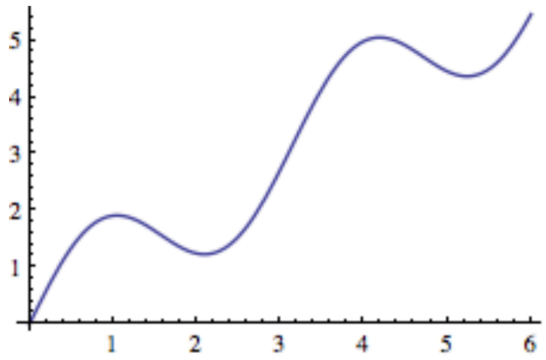
1. Choose a partition of $[1,5]$ into 5 subintervals of equal length $\Delta x = \frac{5-1}{5} = 0.8$. Choose x_i^* to be the left endpoint of each subinterval.



2. Choose again a partition of $[1,5]$ into 5 subintervals of equal length $\Delta x = \frac{5-1}{5} = 0.8$. Now choose x_i^* to be the right endpoint of each subinterval.



3. Choose again a partition of $[1,5]$ into 5 subintervals of equal length $\Delta x = \frac{5-1}{5} = 0.8$. Now choose x_i^* to be the midpoint of each subinterval.



4. Now choose a partition consisting of 5 subintervals of unequal length. Choose x_i^* to be an arbitrary point of each subinterval.

